

# Jaehyeon Moon

Yonsei University

Combined M.S. & Ph.D. student

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**Website**

<https://github.com/JaeHyeonMoon>

**Location**

50, Yonsei-ro, Seodaemun-gu, Seoul, Korea, 03722

## Research Interests

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- Network quantization
- Generative models

## Education

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Mar. 2020 – Present      **Yonsei University**  
*School of Electrical and Electronic Engineering*  
*Seoul, Korea*  
Advisor: Bumsub Ham  
Combined M.S. & Ph.D. student  
GPA: 3.93 / 4.30

Mar. 2016 – Feb. 2020      **Yonsei University**  
*School of Electrical and Electronic Engineering*  
*Seoul, Korea*  
Advisor: Sangyoun Lee  
B.S. in Electrical and Electronic Engineering  
GPA: 4.06 / 4.30 (Magna Cum Laude)

## Publications

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- **Jaehyeon Moon**, Bumsub Ham, "Shift-and-Sum Quantization for Visual Autoregressive Models", ICLR, 2026 (submitted).
- Dohyung Kim\*, **Jaehyeon Moon**\*, Junghyup Lee, Geon Lee, Jeimin Jeon, Bumsub Ham, "Token-Based Dynamic Bit-Width Assignment for ViT Quantization", Pattern Recognition, 2025.  
\*Equal contribution.
- **Jaehyeon Moon**, Dohyung Kim, Junyong Cheon, Bumsub Ham, "Instance-Aware Group Quantization for Vision Transformers", CVPR, 2024.
- Seunghoon Lee\*, Jeongwoo Choi\*, Byunggwon Son, **Jaehyeon Moon**, Jeimin Jeon, Bumsub Ham, "AccuQuant: Simulating Multiple Denoising steps for Quantizing Diffusion Models", NeurIPS, 2025.  
\*Equal contribution.
- Dohyung Kim, Junghyup Lee, Jeimin Jeon, **Jaehyeon Moon**, Bumsub Ham, "Toward INT4 Fixed-Point Training via Exploring Quantization Error for Gradients", ECCV, 2024.

## Experiences

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Jul. 2023 – **TIPS project**  
Jun. 2026 *Research engineer, Articron*

Developed pruning and quantization algorithms to deploy deep neural networks on compute-in-memory (CIM) chips under the TIPS startup R&D program

## Patents

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- 함범섭, **문재현**, 김도형, 전준용, "인공 신경망을 위한 양자화 장치와 방법 및 이를 구비하는 이미지 처리 장치", US, 18/456,020
- 함범섭, 김도형, 이중협, **문재현**, 이건, 전제민, "비전 트랜스포머를 위한 토큰 양자화 장치 및 방법", KR-Application No. 10-2024-0137421
- 함범섭, 김도형, 이중협, 전제민, **문재현**, "인공 신경망을 위한 양자화 장치 및 방법", KR-Application No. 10-2023-0116857
- 함범섭, **문재현**, 김도형, 전준용, "인공 신경망을 위한 양자화 장치와 방법 및 이를 구비하는 이미지 처리 장치", KR-Application No. 10-2023-0072090

## Awards & Scholarships

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- 국가우수장학금(이공계), 과학기술정보통신부, Republic of Korea (Sep. 2018 – Feb. 2020)